

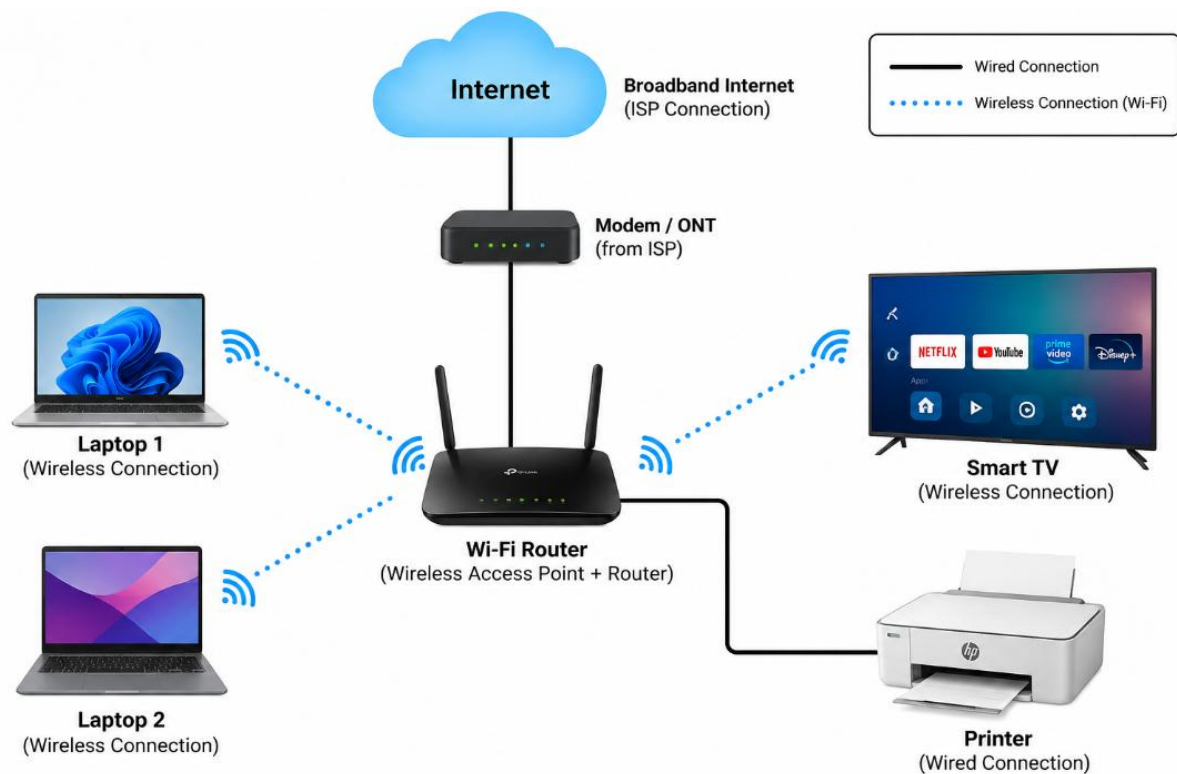
Build a SOHO Network

Q.1 Draw a network diagram for a SOHO setup

Solution: SOHO Network Diagram

A small office/home office network can be created using one broadband Internet connection, a Wi-Fi router, two laptops, one smart TV, and one printer.

For this setup, the laptops, smart TV, and printer can connect wirelessly, while the broadband connection uses an Ethernet cable to the router.



Connection Types

Device	Connection	Example IP
Broadband/ISP	Wired Ethernet	Public IP
Wi-Fi Router	Main network device	192.168.10.1
Laptop 1	Wireless	DHCP
Laptop 2	Wireless	DHCP
Smart TV	Wireless	192.168.10.20

Device	Connection	Example IP
Printer	Wireless	192.168.10.10

The subnetting practical provided earlier uses **192.168.10.0/24**, with **192.168.10.1** assigned to the router and other addresses assigned to network devices.

The Ethernet practical also shows the router being used to connect network devices and an ISP/ONT, which is consistent with this SOHO arrangement.

Q.2 List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Solution: Required Hardware

For this small SOHO network, I would use the following hardware.

No.	Hardware	Suggested Model	Quantity	Purpose
1	Broadband/ONT modem	ISP-provided ONT/modem	1	Provides Internet connection
2	Wi-Fi Router	TP-Link Archer C6 AC1200	1	Provides Wi-Fi, DHCP and NAT
3	Laptop	Existing laptop	2	Wireless client devices
4	Smart TV	Any Wi-Fi-enabled Smart TV	1	Wireless entertainment device
5	Printer	Epson EcoTank L3250	1	Network printer
6	Ethernet cable	Cat6 RJ-45	1–2	ONT → router and optional wired devices
7	Power adapters	Supplied with devices	As required	Power
8	Network switch	TP-Link TL-SG108E	Optional	Additional wired devices
9	Wi-Fi extender	TP-Link RE315	Optional	Extends Wi-Fi coverage

Q.3 Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Solution: Router Configuration

I would configure the router with a unique Wi-Fi name, WPA2 security, a strong Wi-Fi password, and a new administrator password.

Step 1: Connect the Router

1. Connect the broadband/ONT cable to the router's WAN/Internet port.
2. Connect the power adapter and turn on the router.
3. Connect a laptop to the router using Wi-Fi or Ethernet.

Step 2: Open Router Settings

1. Open a web browser.
2. Enter the gateway address, for example 192.168.10.1.
3. Log in using the administrator credentials.

Step 3: Configure Wi-Fi

Go to Wireless/Wi-Fi Settings and set:

- SSID: SOHO_Network_2026
- Security: WPA2-Personal
- Encryption: AES
- Wi-Fi Password: Use a strong unique password.

Step 4: Change Admin Password

Go to Administration/System Tools → Password and replace the default administrator password with a strong unique password.

Step 5: Enable DHCP

Enable the DHCP server with:

- Router IP: 192.168.10.1
- Subnet Mask: 255.255.255.0
- DHCP Range: 192.168.10.100 – 192.168.10.199

DHCP automatically assigns IP addresses to connected devices within the network.

Q.4 Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Solution: IP Address Plan

I used the network:

Network: 192.168.10.0/24

Subnet Mask: 255.255.255.0

Router: 192.168.10.1

I kept the static devices outside the DHCP range to avoid IP-address conflicts.

DHCP Range : 192.168.10.100 – 192.168.10.199

IP Address Table

The MAC addresses below are **made-up sample MAC addresses**

Device	MAC Address	Connection	IP Address	Method
Wi-Fi Router	00:11:22:33:44:01	Wired/Wi-Fi	192.168.10.1	Static
Printer	00:11:22:33:44:10	Wireless	192.168.10.10	Static
Smart TV	00:11:22:33:44:20	Wireless	192.168.10.20	Static
Laptop 1	00:11:22:33:44:31	Wireless	192.168.10.101	DHCP
Laptop 2	00:11:22:33:44:32	Wireless	192.168.10.102	DHCP

Static IP Devices

Printer:

IP Address: 192.168.10.10

Subnet Mask: 255.255.255.0

Gateway: 192.168.10.1

DNS: 8.8.8.8

Smart TV:

IP Address: 192.168.10.20

Subnet Mask: 255.255.255.0

Gateway: 192.168.10.1

DNS: 8.8.8.8

Laptops

The laptops use DHCP, so the router automatically provides their IP addresses.

For example:

Laptop 1 → 192.168.10.101

Laptop 2 → 192.168.10.102

The subnetting practical similarly assigns individual addresses to devices while keeping the network and broadcast addresses reserved.

Q.5 Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Solution: Troubleshooting Steps

If one laptop cannot access the Internet while other devices are working, I would troubleshoot it step by step.

1. Check IP Address

Open Command Prompt and run : ipconfig

Check the IPv4 address, subnet mask, and default gateway.

Example:

- IPv4: 192.168.10.101
- Subnet Mask: 255.255.255.0
- Gateway: 192.168.10.1

If the address is incorrect, run : ipconfig /release

2. Test Router Connection

Run : ping 192.168.10.1

If it fails, reconnect to Wi-Fi, check the password and signal, or restart the router.

3. Check DNS

Run : nslookup www.google.com

If DNS fails, try using 8.8.8.8 as the DNS server.

4. Test Internet Connection

Run : ping 8.8.8.8

If the router responds but 8.8.8.8 does not, check the router's WAN/Internet status, ONT/modem, and ISP connection.